

ST PETERSBURG WHITTED, FL, USA

WMO: 722104

Lat: 27.765N

Lon: 82.628W

Elev: 8

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 92806

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.9	48.0	28.6	22.6	52.8	32.5	27.0	52.9	24.2	58.2	21.9	56.6	11.2	0	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.0	91.9	77.4	90.6	77.4	89.8	77.5	81.3	85.6	80.4	85.6	79.8	85.6	6.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.2	83.5	79.1	150.8	83.4	78.1	145.9	83.3	44.7	85.8	43.8	85.5	43.1	85.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	19.7	18.0	37.7	94.9	5.0	1.8	34.2	96.2	31.3	97.2	28.5	98.3	24.9	99.6		
			33.7	82.2	3.6	1.9	31.1	83.6	29.0	84.6	27.0	85.7	24.4	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.8	61.9	65.0	69.1	74.3	79.3	82.9	84.0	84.3	82.8	78.0	70.3	65.6
	DBStd	9.16	7.32	6.43	5.58	4.18	3.20	2.45	2.33	2.23	2.51	4.67	5.58	6.57
	HDD50	11	8	1	0	0	0	0	0	0	0	0	0	2
	HDD65	349	144	74	30	3	0	0	0	0	0	2	21	75
	CDD50	9077	376	420	591	730	909	986	1054	1063	984	869	608	487
	CDD65	3940	47	74	156	283	444	536	589	598	534	405	179	95
	CDH74	35927	73	163	510	1517	3767	5928	6985	7147	5755	3255	629	198
	CDH80	11914	1	5	41	244	1051	2144	2771	2886	1933	765	65	8
Wind	WSAvg	8.8	8.4	8.4	9.3	9.5	9.3	8.4	7.5	7.9	9.5	10.1	9.3	8.4
Precipitation	PrecAvg	50.2	2.5	2.7	3.0	2.4	2.5	6.9	8.4	8.6	6.7	2.6	1.5	2.4
	PrecMax	69.6	5.5	11.6	9.1	8.6	8.9	14.1	15.2	13.6	13.6	6.6	4.8	15.6
	PrecMin	38.6	0.4	0.4	0.3	0.2	0.2	3.1	4.0	4.6	1.7	0.4	0.2	0.5
	PrecStd	8.0	1.5	2.3	2.4	1.7	2.1	3.0	2.6	2.8	3.1	1.4	1.1	3.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.9	81.0	82.9	86.6	91.0	93.2	93.4	93.4	92.1	90.0	84.3	81.0
		MCWB	69.8	70.6	70.9	72.1	74.3	77.0	77.5	78.2	77.3	76.9	73.4	72.3
	2%	DB	76.0	78.6	81.3	84.4	88.5	91.0	91.3	91.5	90.3	87.8	81.7	78.6
		MCWB	68.2	69.5	70.1	71.5	73.5	76.8	77.7	78.1	77.2	76.3	72.6	71.0
	5%	DB	73.5	76.0	79.2	82.6	87.1	89.7	90.1	90.3	89.0	85.8	79.7	76.2
		MCWB	67.4	68.1	68.8	71.1	73.1	76.9	77.9	78.3	77.1	75.4	71.6	70.2
	10%	DB	71.5	73.5	77.0	81.2	85.2	88.2	88.7	89.1	87.8	84.0	77.7	73.9
		MCWB	66.0	67.0	68.0	70.5	72.6	76.6	77.8	78.3	77.1	74.9	70.2	69.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.2	73.0	74.9	77.2	78.0	82.0	82.0	82.4	81.7	80.4	76.8	74.3
		MCDB	75.9	77.8	80.0	82.1	83.6	84.1	86.9	86.2	85.3	83.2	80.5	78.1
	2%	WB	70.2	71.3	73.1	75.0	76.7	80.1	80.7	81.3	80.3	78.4	75.3	72.8
		MCDB	74.0	75.4	77.6	80.4	83.2	85.1	86.1	86.2	84.9	83.5	79.2	76.5
	5%	WB	68.6	70.0	71.7	73.7	76.0	79.0	80.0	80.4	79.4	77.6	73.6	71.4
		MCDB	72.4	74.3	75.8	79.2	82.5	85.3	85.8	86.1	84.8	83.3	77.8	74.8
	10%	WB	66.8	68.7	70.2	72.6	75.2	78.2	79.3	79.7	78.5	76.8	72.0	70.0
		MCDB	70.5	72.8	74.9	78.4	81.8	84.8	85.5	86.0	84.1	82.4	76.4	73.2
Mean Daily Temperature Range	5% DB	MDBR	12.9	12.8	12.7	12.5	12.0	10.8	10.1	10.0	9.9	10.4	11.6	11.9
		MCDBR	13.8	13.8	14.0	13.6	13.5	12.4	11.5	11.3	11.3	10.9	12.1	12.5
		MCWBR	7.9	7.3	6.6	5.7	5.3	4.7	4.7	4.5	4.4	4.3	6.4	6.5
	5% WB	MCDBR	12.9	12.3	11.8	12.1	11.7	10.9	10.4	10.1	10.0	10.0	10.8	11.6
	MCWBR	9.1	7.7	7.4	6.8	5.3	5.1	5.1	4.9	4.7	4.5	6.9	7.6	
Clear-Sky Solar Irradiance	taub	0.343	0.356	0.369	0.394	0.424	0.448	0.472	0.452	0.423	0.390	0.355	0.344	
	taud	2.528	2.498	2.447	2.340	2.315	2.329	2.268	2.336	2.383	2.461	2.505	2.549	
	Ebn at Noon	286	291	292	286	276	268	261	266	272	276	279	281	
	Edh at Noon	29	32	35	41	42	41	43	40	37	33	29	27	
All-Sky Solar Radiation	RadAvg	1129	1366	1726	2069	2195	1988	1885	1809	1643	1518	1235	1045	
	RadStd	82	90	143	98	117	160	113	112	109	93	65	86	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157
(47)	Regional (32 neighbors)	+0.77	+1.44	N/A	N/A	N/A	N/A	-5	-81	+311	+218

Nomenclature: See separate page